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Prediction of available soil nitrogen using Vis–NIR and MIR spectroscopy coupled with chemometric and machine learning approaches in inceptisols of India

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ABSTRACT

Soil nitrogen is a fundamental indicator of soil fertility and crop productivity, and its rapid assessment is essential for efficient land management and sustainable agricultural practices. This study investigates the potential of Vis–NIR and MIR spectroscopy for predicting available soil nitrogen collected from the intensively cultivated Indo-Gangetic Plains of India. Soil samples were analyzed using the standard laboratory method and corresponding spectral signatures were processed using multiple preprocessing techniques, among which Standard Normal Variate (SNV) showed the greatest enhancement in model performance. Four regression algorithms Partial Least Squares Regression (PLSR), Random Forest (RF), Support Vector Regression (SVR) and Multivariate Adaptive Regression Splines (MARS) were calibrated for both the Vis–NIR and MIR spectral domains. Model evaluation demonstrated that MIR spectroscopy combined with PLSR yielded the highest prediction accuracy (R^2 = 0.84, RMSE = 9.95, RPD = 2.70), outperforming all Vis-NIR models and other machine-learning approaches. RF, MARS and SVR exhibited comparatively lower performance, with RPD values between 1.06 and 1.66. Based on the results, MIR spectroscopy combined with the PLSR algorithm is recommended as a rapid and reliable approach for estimating available nitrogen in Inceptisol soils of the Indo-Gangetic Plains. Adopting this technique can enhance nitrogen management while reducing time, cost and waste for sustainable soil fertility monitoring.

1. Introduction

Soil plays a vital role in regulating global nutrient and elemental cycles particularly the nitrogen (N) cycle, which regulates ecosystem productivity, nutrient dynamics, and greenhouse gas fluxes. Quantification of soil nitrogen is essential to understand nutrient transformations, gaseous emissions, and soil fertility dynamics under changing climatic conditions and for promoting sustainable agriculture (Zhao et al., 2022). Nitrogen is also critical for developing soil management strategies that enhance soil health and productivity to meet the increasing global demand for food and bioresource production (McBride et al., 2011; Liu et al., 2021). Owing to continuous shifts in land use patterns, cropping systems, and agricultural practices, periodic assessment of soil nitrogen concentrations is essential for sustainable nutrient management (Dinakaran et al., 2016).

Traditional laboratory methods for estimating soil nitrogen, although precise, involve high costs and lengthy procedures, limiting their applicability for large-scale or rapid evaluations (McDowell et al., 2012; Demattê et al., 2019). Reflectance spectroscopy in the visible–near infrared (Vis–NIR) and mid-infrared (MIR) regions has emerged as a rapid, non-destructive, and economical alternative for soil analysis. These techniques require minimal sample preparation and can estimate multiple soil properties simultaneously from a single spectral measurement (Viscarra Rossel et al., 2006; Stenberg et al., 2010; Lima

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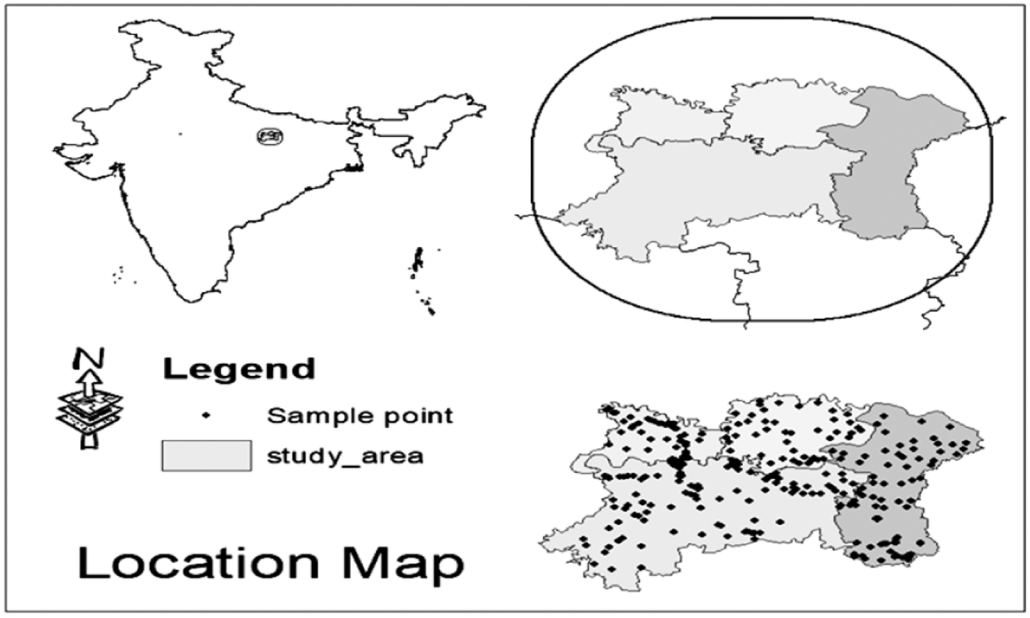


Fig. 1. Location Map of study area.

et al., 2024). The spectral characteristics in these regions reflect the composition of soils through the overtones and combination bands of fundamental molecular vibrations associated with functional groups such as O–H, C═O, N–H, and C–H in organic and inorganic compounds (Zhang et al., 2024).

Spectroscopy-based nitrogen estimation is instrumental in advancing precision agriculture, assessing soil fertility, and promoting climate-smart nutrient management (Wang et al., 2024 and Tang et al., 2024). Several studies in tropical and subtropical regions have successfully predicted total or organic nitrogen using Vis–NIR or MIR spectroscopy (Dotto et al., 2018; Zhao et al., 2022), with MIR generally providing higher predictive accuracy due to stronger absorption of fundamental molecular vibrations (Bellon-Maurel and McBratney, 2011; Vohland et al., 2014). However, prediction accuracy depends strongly on pre-processing and multivariate calibration techniques.

Spectral preprocessing methods such as Standard Normal Variate (SNV), first and second derivatives, and smoothing are critical for removing noise and enhancing meaningful features before modeling (Rosin et al., 2020). Advanced chemometric and machine learning algorithms Partial Least Squares Regression (PLSR), Random Forest (RF), Support Vector Regression (SVR), and Multivariate Adaptive Regression Splines (MARS) are increasingly applied to establish robust relationships between spectral data and soil reference values (Ben-Dor and Banin, 1995; Felicísimo et al., 2013; Seema et al., 2025). Despite this progress, there is still no standardized combination of spectral range, pre-processing, and modeling approach suitable for all soil types and regions (Stenberg et al., 2010; Shi et al., 2024).

The Indo-Gangetic Plains (IGP) of India constitute one of the most intensively cultivated and agriculturally important regions globally, characterized by deep alluvial soils, diverse cropping systems, and variable climatic conditions ranging from semi-arid in the west to sub-humid in the east (Pal et al., 2009 and Mandal et al., 2025). The region experiences high spatial variability in soil fertility due to intensive land use, irrigation, and fertilizer applications (Mandal et al., 2025). However, very few studies have evaluated the potential of Vis–NIR and MIR spectroscopy for available nitrogen prediction in Indo-Gangetic soils, with most prior work focusing on total or organic N estimation. Given the strong spatial variability in soil properties and the limited availability of reliable nitrogen prediction models for this region, there is a clear need for region-specific calibration and validation of spectral models using appropriate preprocessing and chemometric techniques. The present study aims to (i) evaluate the potential of Vis–NIR and MIR spectroscopy for predicting available soil nitrogen, (ii) assess the effects of different preprocessing techniques, and (iii) compare the predictive performance of PLSR, RF, SVR, and MARS models. It is hypothesized that MIR spectroscopy, combined with suitable preprocessing and machine learning algorithms, will offer superior predictive accuracy for available nitrogen in the Inceptisols of the Indo-Gangetic Plains of India.

2. Materials and methods

2.1. Study area and soil sampling

The investigation was carried out in the middle Indo-Gangetic Plains of Uttar Pradesh, India, covering the districts of Varanasi, Chandauli, Sant Ravidas Nagar, and Mirzapur (82°30' and 83°30' E; 24°30'and 25°30' N), occupying nearly 9604 square km (Fig. 1).

The region is characterized by a semi-arid to sub-humid climate, with hot summers where temperatures may reach 45 °C and cold winters with temperatures falling below 10 °C. The mean summer, winter, and soil temperatures are 33.1, 18.8, and 25.6 °C, respectively. Annual rainfall ranges from 970 to 1100 mm, of which nearly 80 % occurs during the monsoon period from June to September. The soils of the region are predominantly Inceptisols and alluvial soils, developed from alluvium and sandstone parent materials. Overall, the area contains alluvial and residual soils dominated by illite and related clay minerals, with textures ranging from sandy loam to clay. The region predominantly follows a rice-based cropping system, with wheat as the major winter crop. A total of 280 geo-referenced composite randomly selected soil samples were collected in October 2018 from soil surface (0–15 cm) after crop harvest using a stainless-steel auger. Sampling coordinates were recorded with a handheld Global Positioning System (Garmin Etrex). The samples were air-dried, gently ground with a wooden mortar and pestle, and sieved through a 2 mm mesh. Available nitrogen in soil was estimated using the alkaline permanganate method described by Subbiah and Asija (1956), which is widely adopted for its speed, reliability, and suitability for routine soil testing. In this procedure, 5 g of soil was transferred to a distillation tube, followed by the addition of 25 mL of 0.32 % $KMnO_4$ solution. A 250 mL conical flask containing 20 mL of 2 % boric acid and mixed indicator was placed at the receiving end of the condenser. Subsequently, 25 mL of 2.5 % NaOH was added to the distillation tube, and distillation was carried out for 9 min. During distillation, the

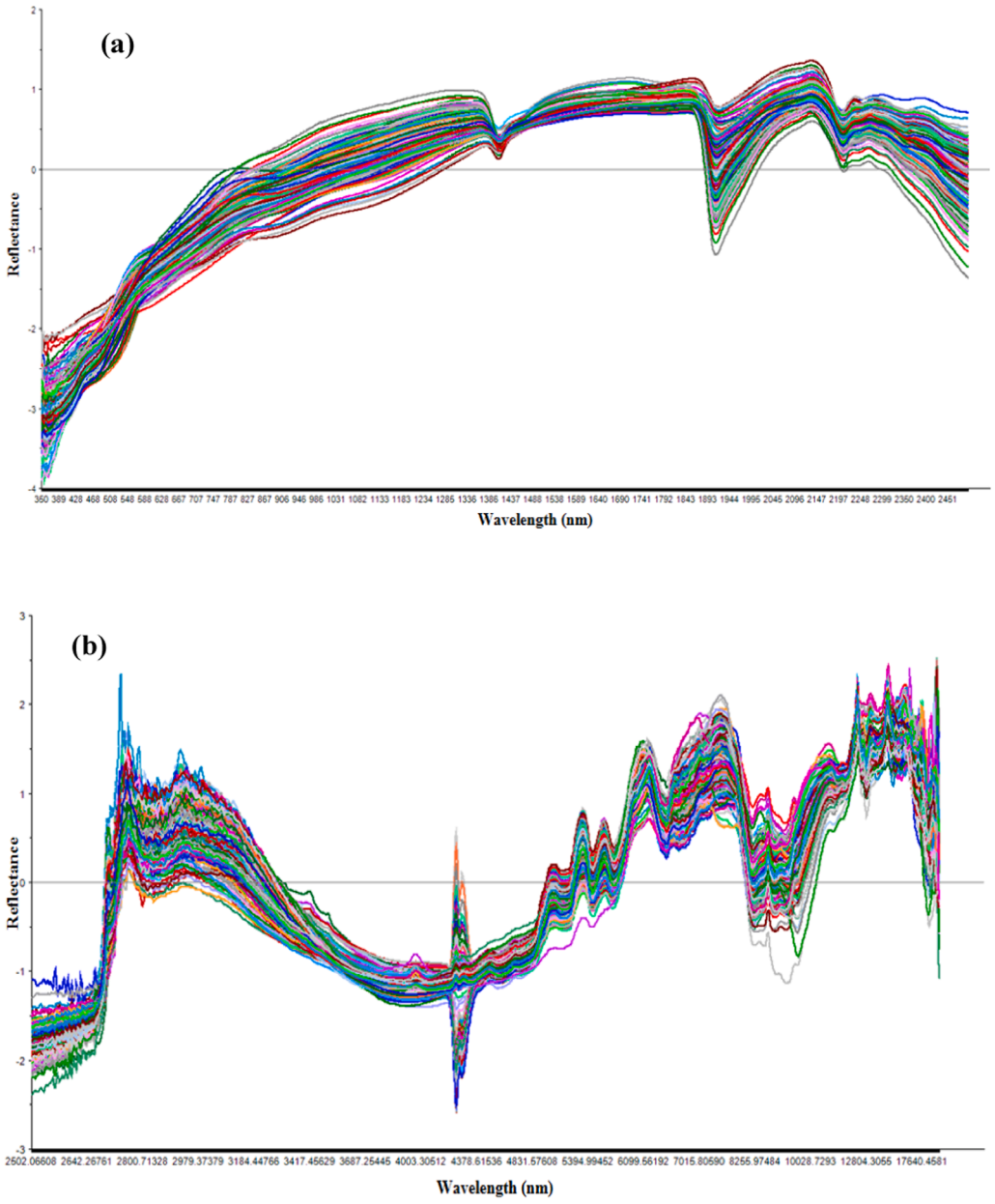


Fig. 2. Preprocessing standard normal variate (SNV) of the spectral curves for all soil samples a) NIR spectral data b) MIR spectral data.

Table 1
Descriptive statistics of soil Available Nitrogen (kg ha^{-1}).

Dataset	N	Range	Mean	Median	SD	CV	Skewness	Kurtosis
All	280	87.80–288.51	191.38	200.70	27.26	5.71	-0.30	0.89
Calibration	196	87.80–288.51	188.94	188.16	34.69	5.43	-0.24	0.69
Validation	84	112.89–263.42	199.44	200.72	27.28	7.31	-0.20	1.87

released ammonia (NH_3) was absorbed in the boric acid solution, producing a green coloration. The distillate was then titrated against 0.02 N H_2SO_4 to a pink endpoint, and the available nitrogen content was calculated and expressed as kg ha^{-1} (Ghosh et al., 2012).

For spectral analysis, the processed soil was homogenized using a Retsch Mortar Grinder RM 200 for 3 min to achieve a uniform 0.5 mm particle size, ensuring homogeneity and analytical precision. Finely ground samples were used for mid-infrared (MIR) spectral analysis, whereas 2 mm sieved samples were employed for visible–near infrared (Vis–NIR) measurements (Ghosh et al., 2020; 2022 and Seema et al., 2025). For MIR spectroscopy, soil samples were finely ground to 0.5 mm to minimize particle-size–induced scattering and to obtain clear, well-defined absorption features, whereas Vis–NIR measurements were performed on 2 mm sieved samples because this wavelength region is less sensitive to particle size and requires preserving natural soil reflectance characteristics.

Table 2
Calibration and validation statistics for visual- near infrared (Vis-NIR) region employed for quantifying soil Available Nitrogen.

Spectral preprocessing[a]	Model[b]	Calibration		Validation		[c]RPD
		R^2	RMSE (kg ha^{-1})	R^2	RMSE (kg ha^{-1})	
Raw	PLS	0.57	34.90	0.55	26.78	1.30
	RF	0.59	32.33	0.52	28.26	1.13
	MARS	0.44	38.37	0.45	30.78	1.19
	SVR	0.42	38.68	0.53	28.82	1.09
SNV	PLS	0.95	7.33	0.81	12.26	2.16
	RF	0.87	11.90	0.68	16.78	1.54
	MARS	0.84	12.37	0.65	17.78	1.52
	SVR	0.82	22.68	0.63	18.72	1.38
MSC	PLS	0.82	14.41	0.54	20.26	1.56
	RF	0.77	15.92	0.51	21.78	1.14
	MARS	0.70	16.87	0.50	23.78	1.32
	SVR	0.73	17.69	0.53	24.82	1.08
ABS	PLS	0.63	25.32	0.46	28.53	0.88
	RF	0.70	20.38	0.54	26.46	1.08
	MARS	0.64	23.36	0.48	20.31	0.98
	SVR	0.60	23.32	0.41	23.85	0.85
SG-First order derivative	PLS	0.64	25.92	0.46	28.78	0.88
	RF	0.66	25.34	0.43	22.06	0.81
	MARS	0.63	22.38	0.38	27.88	0.85
	SVR	0.59	28.62	0.40	26.42	0.92
SG-Second order derivative	PLS	0.55	34.90	0.55	23.78	1.07
	RF	0.54	32.33	0.52	24.26	1.04
	MARS	0.51	38.37	0.44	30.08	0.86
	SVR	0.50	38.68	0.53	24.82	0.95

ABS, log(1/reflectance).
[b]PLS,Partial Least Square Regression; RF, Random Forest Regression, SVR, Support Vector Regression; MARS, Multivariate Adaptive Regression Splines.
[c]RPD,residual prediction deviation
[a] Raw, Reflectance; SG, Savitzky-Golay; MSC, Multiplicative scatter correction; SNV, Standard normal variate;

Table 3
Calibration and validation statistics for mid infra-red (MIR) range employed for quantifying soil Available Nitrogen.

Spectral preprocessin[a]	Model[b]	Calibration		Validation		RPD[c]
		R^2	RMSE (kg ha^{-1})	R^2	RMSE (kg ha^{-1})	
Raw	PLS	0.50	24.34	0.43	26.15	0.66
	RF	0.60	21.34	0.40	26.05	0.63
	MARS	0.51	24.13	0.46	26.94	0.52
	SVR	0.52	24.03	0.44	26.34	0.56
SNV	PLS	0.90	10.34	0.84	9.95	2.70
	RF	0.80	15.34	0.56	16.15	1.66
	MARS	0.81	17.03	0.46	22.94	1.17
	SVR	0.79	19.03	0.44	25.34	1.06
MSC	PLS	0.80	15.44	0.70	12.95	2.20
	RF	0.70	20.34	0.54	23.15	1.22
	MARS	0.78	18.43	0.44	23.04	1.12
	SVR	0.71	20.23	0.40	26.34	1.01
ABS	PLS	0.66	22.31	0.48	26.12	0.69
	RF	0.71	20.30	0.61	22.25	0.82
	MARS	0.61	23.13	0.52	25.44	0.51
	SVR	0.63	23.21	0.58	25.34	0.58
SG-First order derivative	PLS	0.61	23.04	0.48	25.15	0.68
	RF	0.64	20.34	0.51	25.85	0.61
	MARS	0.50	24.53	0.46	25.24	0.52
	SVR	0.54	23.43	0.40	26.02	0.56
SG-Second order derivative	PLS	0.56	23.32	0.46	25.14	0.68
	RF	0.68	20.37	0.48	25.65	0.71
	MARS	0.52	23.17	0.44	26.04	0.53
	SVR	0.58	23.13	0.46	25.38	0.61

[b]PLS,Partial Least Square Regression; RF, Random Forest Regression, SVR, Support Vector Regression; MARS, Multivariate Adaptive Regression Splines.
[c]RPD,residual prediction deviation
[a] Raw, Reflectance; SG, Savitzky-Golay; MSC, Multiplicative scatter correction; SNV, Standard normal variate; ABS, log(1/reflectance).

2.2. *Spectroscopic measurement and pre-processing of spectra*

2.2.1. *Collection of soil spectral data in Vis-NIR range*

Soil reflectance spectra were acquired using a FieldSpec 3 FR spectroradiometer (Analytical Spectral Devices Inc., USA) equipped with a 10 mm contact probe. The instrument operates across the 350–2500 nm range, encompassing the VIS, NIR, and SWIR regions. It provides a spectral resolution of 3 nm at 700 nm and 10 nm at 2100 nm, with a uniform sampling interval of 1.0 nm across the 350–2500 nm range, resulting in a total of 2151 spectral bands. About 50 g of soil was placed in a 10 cm-diameter container and the surface leveled with a rubber cork (Mouazen et al., 2010). Four spectra one per quadrant were recorded and averaged to represent each sample. For each soil sample, a reference spectrum was also collected using a 9.2-cm diameter Spectralon white reference panel (Labsphere) (Ghosh et al., 2020 and Seema et al., 2025).

2.2.2. *Collection of soil spectral data in MIR range*

Air-dried and homogenized (0.5 mm) soil samples were loaded into sample cups of a Bruker Alpha FT-MIR Spectrometer. The instrument was allowed to stabilize for approximately 2 h, after which a background scan was collected using the standard gold cap provided with the spectrometer. Background scans were repeated at 30 min intervals during sample scanning to account for atmospheric and instrumental drift and to improve the signal-to-noise ratio. Following calibration and background correction, MIR spectra were acquired across the 4000–400 cm^{-1} range (equivalent to 2500–25,000 nm) (Ghosh et al., 2022 and Seema et al., 2025).

The raw soil spectra were collected from the ASD and OPUS file from ASD FieldSpec® and MIR-spectrometer respectively. Subsequently, it was pre-processed through R statistical software using the "prospectr" package. Spectral pre-processing was applied to both the Vis–NIR and MIR datasets to minimize scattering and enhance chemically relevant features, was carried out in R software (Rinnan et al., 2009). Six mathematical treatments (1) reflectance, (2) log(1/reflectance) (absorbance), (3) first derivative (FD), (4) second derivative (SD) with Savitzky–Golay filter, (5) standard normal variate (SNV), and (6) multiplicative scatter correction (MSC) was applied. The Savitzky–Golay filter used a third-order polynomial with a window size of 25 points was used to correct baseline variations, while SNV and MSC eliminated particle-size and additive multiplicative scattering effects (Dotto et al., 2018; Ghosh et al., 2020; 2022 and Seema et al., 2025).

2.3. ***Multivariate regression models***

Regression analysis was conducted in R Software version 3.3.3 (The R Development Core Team, 2017) to predict soil nitrogen using four algorithms: Partial Least Squares Regression (PLSR), Random Forest (RF), Support Vector Regression (SVR), and Multivariate Adaptive Regression Splines (MARS). For model development, hyperparameters were optimized using a grid-search strategy with 10-fold cross-validation. All models were trained and validated using the same cross-validation approach to ensure robustness and prevent overfitting. Cross-validation was used to calibrate the spectral data with the wet chemistry/laboratory soil data to obtain predictive models for each soil property. This is one of the most frequently used techniques for chemometric modeling and it can be considered the de facto standard method in soil spectroscopy (Wijewardane et al., 2018). The respective R packages 'pls', 'randomForest', 'kernlab', and 'earth, plotmo, plotrix, Teaching Demos' were used (Clyde, 2017).

Partial Least Squares (PLS) regression is a multivariate statistical technique that extracts latent components that best explain the variation in the response variable (Y). These components are then used to construct a reduced set of predictors (X), effectively lowering

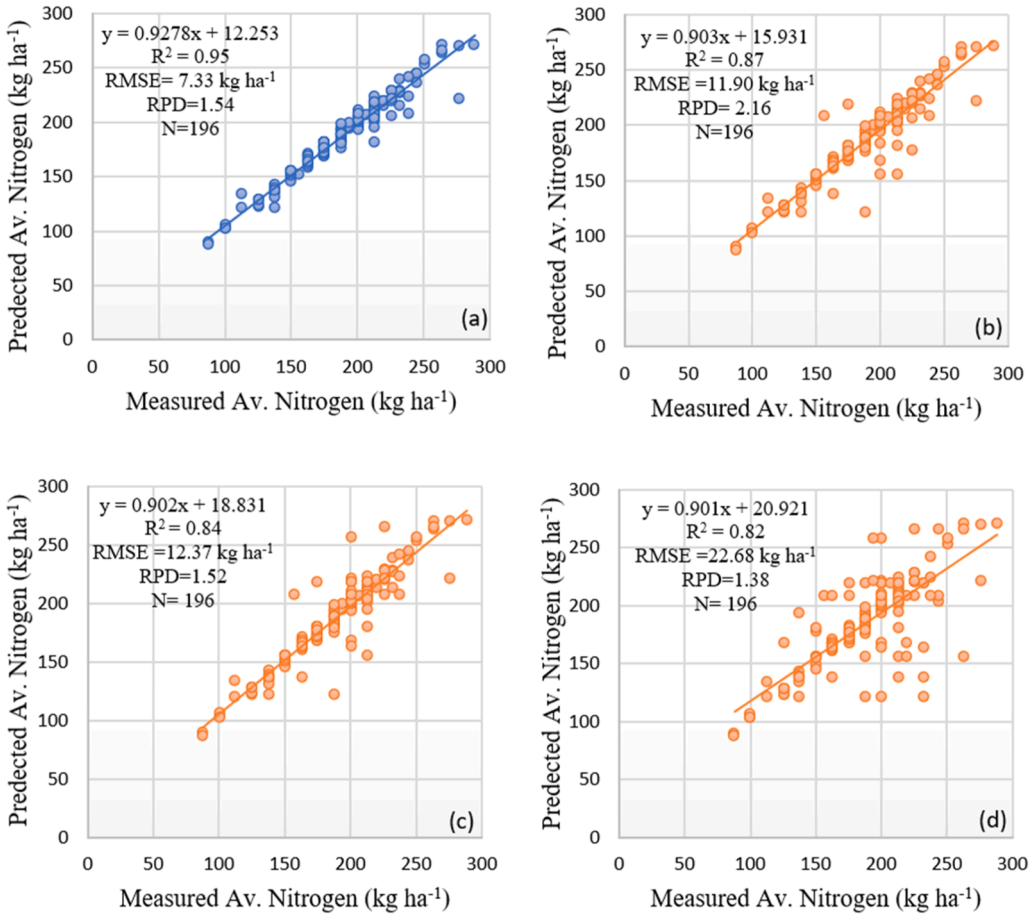


Fig. 3. Near-infrared (NIR) prediction of soil Available Nitrogen calibration datasets of (a) PLSR (b) RF (c) MARS & (d) SVR Models Respectively.

dimensionality, minimizing noise, and mitigating multicollinearity (Wold et al., 2001).

Random Forest (RF) is an ensemble learning algorithm that aggregates the outputs of multiple decision trees to produce a final prediction (Breiman, 2001). Each tree is generated using bootstrap samples of the calibration dataset, and at every node, a random subset of predictors (mtry) is selected for splitting. The final model output represents the mean prediction across all trees. Optimal values of mtry were determined through grid-based tuning combined with 10-fold cross-validation (Ji et al., 2019).

Support Vector Regression (SVR), developed based on statistical learning theory, aims to enhance prediction generalization using the structural risk minimization framework (Filgueiras et al., 2014). Depending on the loss function, two SVR variants are commonly used: ε-SVR, which relies on the parameters epsilon (margin of tolerated error) and C (penalty for exceeding that margin), and ν-SVR, which uses the parameter ν to specify the minimum fraction of support vectors relative to the total calibration samples (Xu et al., 2019).

Multivariate Adaptive Regression Splines (MARS), introduced by Friedman (1991), is a data-driven non-parametric regression approach that models nonlinear and interaction effects through adaptive basis functions. The algorithm divides the predictor space into multiple intervals (splines) defined by knots points where the slope of the relationship changes and fits piecewise linear regressions within each interval. Each basis function represents either the left or right side of a knot:

$$Y = \max(0, \mathrm{Xf} - \mathrm{cf})$$

$$Y = \max(0, \mathrm{cf} - \mathrm{Xf})$$

where *cf* is the threshold value for a predictor, or knot, and *Xf* denotes a predictor variable.

MARS begins with a forward pass that intentionally produces an overfitted model with a large set of basis functions. This is followed by a backward pruning step that removes functions contributing little to predictive performance, resulting in a parsimonious final model (Friedman, 1991). Model selection is guided by a modified generalized cross-validation (GCV) criterion:

$$GCV = 1n\sum i = 1n[yi - f(xi)]^2[1 - C(m)n]^2$$

where n is the sample size, C(m) quantifies the cost-complexity penalty associated with using m basis functions, and the numerator represents the lack-of-fit measure. This formulation balances flexibility and complexity, ensuring that the optimum model achieves the lowest GCV value while avoiding overfitting.

2.4. Model evaluation

Model performance was evaluated using the coefficient of determination (R^2), root mean square error of prediction (RMSE), and ratio of performance deviation (RPD). Higher RPD values indicated better predictive accuracy. When RPD values were identical, R^2 and RMSE were used to discriminate models. The indices were calculated as:

$$R^2 = 1 - \left(\frac{\sum_i n (Y_{Pred} - Y_{measured})^2}{\sum_i n \quad Y_i - Y_{mean}^{\ 2}} \right) \quad (1)$$

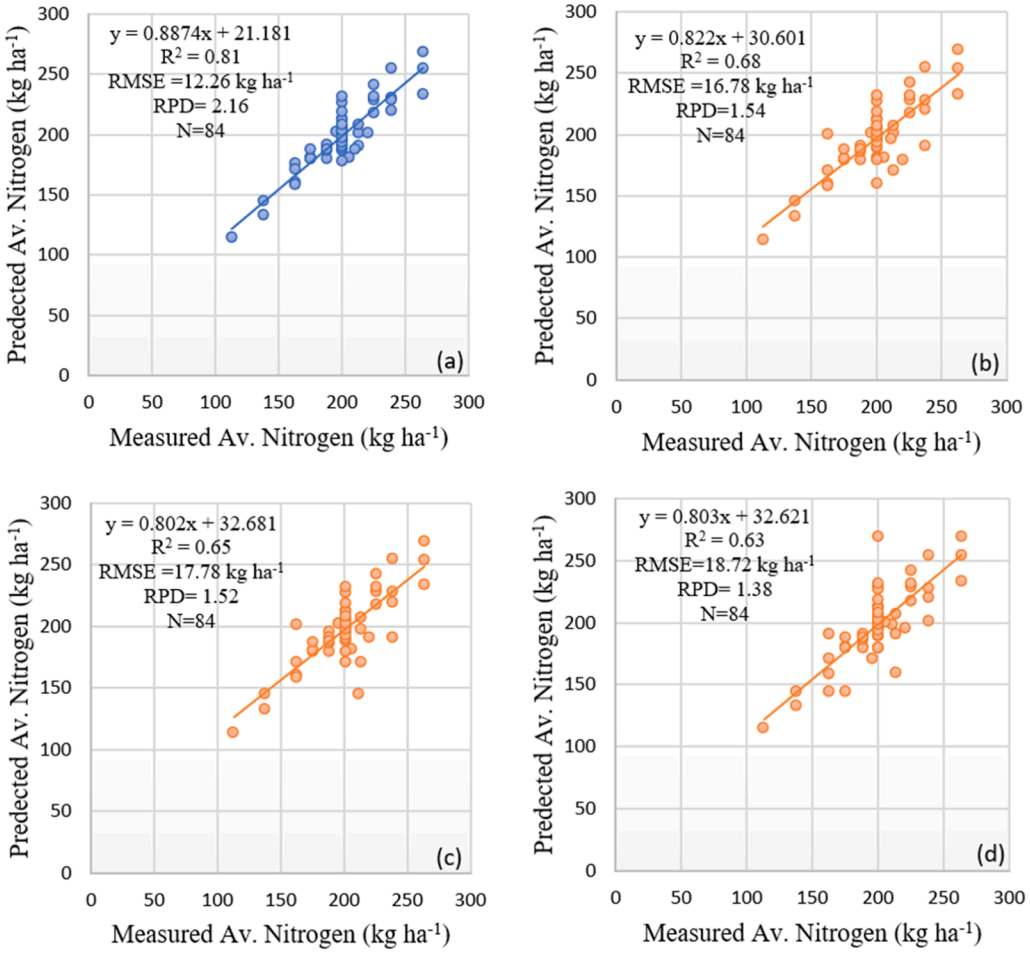


Fig. 4. Near-infrared (NIR) prediction of soil Available Nitrogen validation datasets of (a) PLSR (b) RF (c) MARS & (d) SVR Models Respectively.

$$RMSE = \sqrt{\frac{\sum_i n(Y_{Pred} - Y_{measured})^2}{n-1}} \quad (2)$$

$$RPD = SD_{val.}/RMSE \quad (3)$$

Where, Ypred = predicted values; Ymean = mean of measured values; Y meas = measured values; n = number of predicted or measured values with I = 1,2,…n; SDval= standard deviation of measured values in the validation dataset; and RMSEP = root mean square error of prediction in validation dataset. The Workflow of spectral data processing, model calibration and validation for prediction of available soil nitrogen is depicted in Fig. 2 (Seema et al., 2025).

2.5. Statistical analysis

Descriptive statistics of soil available nitrogen minimum, maximum, mean, standard deviation, coefficient of variation, skewness, and kurtosis were computed using SPSS v23.0.

3. Results

3.1. Descriptive statistics of soil available nitrogen

The descriptive statistics of available nitrogen are presented in Table 1. The full dataset ranged from 87.80 to 288.51 kg ha^{-1} , with a mean of 191.38 kg ha^{-1} , and showed an approximately normal distribution. Both calibration and validation subsets exhibited similar central tendencies and variability, with means of 188.94 and 199.44 kg ha^{-1} , respectively. The soil samples were divided into calibration and validation subsets using a 70:30 ratio, resulting in 196 samples for calibration and 84 samples for validation. The validation range (112.89–263.42 kg ha^{-1}) remained entirely within the calibration range, indicating no extrapolation issues. The moderate CV values (5.43 % and 7.31 %) further confirm that both subsets adequately represent the overall dataset, providing a sound basis for reliable model development and validation.

3.2. Spectral preprocessing influence

The Vis–NIR and MIR spectra were subjected to various preprocessing transformations, and their impact on model performance is summarized in Tables 2 and 3. Among the tested methods, the first derivative, second derivative, and MSC did not substantially improve discrimination of nitrogen-related spectral features compared to raw reflectance spectra. SNV preprocessing produced the highest prediction accuracy across all models, with PLSR + SNV achieving the best performance in both Vis–NIR (R^2 = 0.81, RMSE = 12.26 kg ha^{-1}, RPD = 2.16) and MIR (R^2 = 0.84, RMSE = 9.95 kg ha^{-1}, RPD = 2.70). In comparison, other preprocessing methods such as FD, SD, and MSC showed limited improvement, with lower R^2 values (0.44–0.68) and higher RMSE (16.15–25.34 kg ha^{-1}) in most cases. The SNV-processed spectral curves for all samples are shown in Fig. 2. These results demonstrate that SNV is the most effective preprocessing technique for enhancing nitrogen prediction accuracy, particularly in the MIR region with high RPD value where stronger absorption features contributed to higher predictive performance. The superior performance of SNV preprocessing can be attributed to its ability to reduce variations caused by

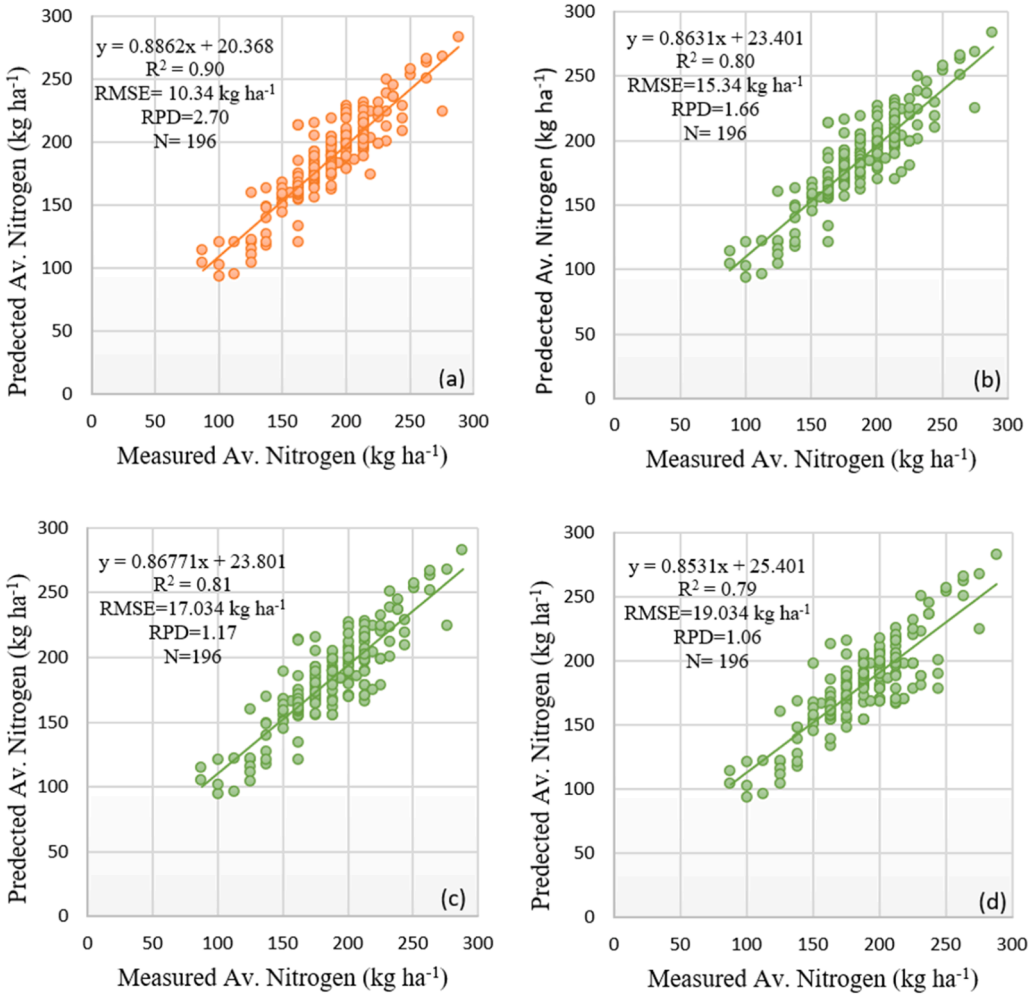


Fig. 5. Mid-infrared (MIR) calibration of soil Available Nitrogen calibration datasets of (a) PLSR (b) RF (c) MARS & (d) SVR Models Respectively.

scattering effects, path length differences, and sensor-related scale and offset discrepancies among MIR spectra. By standardizing the signal intensities, SNV ensured greater spectral consistency across all samples, thereby enabling more reliable extraction of meaningful information into the latent variables of the PLSR model (Barnes et al., 1989; Zou et al., 2025).

Dos Santos (2020) reported that the performance of multivariate models for predicting soil nitrogen exhibited considerable variation in accuracy, largely influenced by differences in pre-processing techniques, spectral regions, and regression algorithms. The reported coefficient of determination (R^2) values ranged from 0.19 to 0.48 g kg^{-1}, with RIPQ values between 1.20 and 3.47. Overall, superior prediction accuracy for nitrogen concentration was achieved using MIR spectra compared with Vis–NIR data. The relatively poor predictive capacity of the Vis–NIR region may be attributed to its weak absorption features and overlapping spectral bands, which result in low signal strength for nitrogen-related chemical compounds (Stenberg et al., 2010)

3.3. Calibrations and predictability of models

Among the four tested regression models, clear differences in predictive performance were observed for both Vis–NIR and MIR datasets. Across calibration datasets, partial least squares regression (PLSR) consistently outperformed random forest (RF), multivariate adaptive regression splines (MARS) and support vector regression (SVR), as reflected by higher coefficients of determination and lower root mean square error (RMSE) values (Tables 2 and 3). Model validation further confirmed the superiority of PLSR, which achieved good predictive performance in the Vis–NIR region (R^2 = 0.81, RMSE = 12.26 kg ha^{-1} and RPD = 2.16) and excellent performance in the MIR region (R^2 = 0.84, RMSE = 9.95 kg ha^{-1} and RPD = 2.70), whereas the remaining models showed only fair to poor prediction accuracy.

The scatter plots of measured versus predicted available nitrogen values for both calibration (Figs. 3 and 5) and validation datasets (Figs. 4 and 6) exhibited strong linear relationships, particularly for the PLSR model, as indicated by high R^2 values and close agreement around the 1:1 line. A comparative evaluation of validation RPD values further revealed consistently higher RPD in the MIR region than in the Vis–NIR region, indicating superior generalization and predictive reliability of MIR-based models. Overall, these results demonstrate that PLSR is the most robust modelling approach for estimating available nitrogen, with MIR spectra providing enhanced sensitivity to nitrogen-related soil constituents compared with the Vis–NIR range.

Buddenbaum and Steffens (2012) reported that derivative-based pre-processing methods tend to increase spectral noise and consequently reduce predictive performance. Therefore, except for the Savitzky Golay derivative (SGDer), other pre-processing techniques may be more appropriate. Although the differences among methods were not statistically significant, the Standard Normal Variate (SNV) transformation is recommended for nitrogen estimation due to its better fitting performance (R^2 = 0.82, RMSE = 0.26 kg ha^{-1}, and RPIQ = 2.69). Similarly, Dos Santos (2020) suggested that the use of MIR spectra, in combination with Partial Least Squares Regression (PLSR) and SNV pre-processing, provides the most reliable approach for predicting

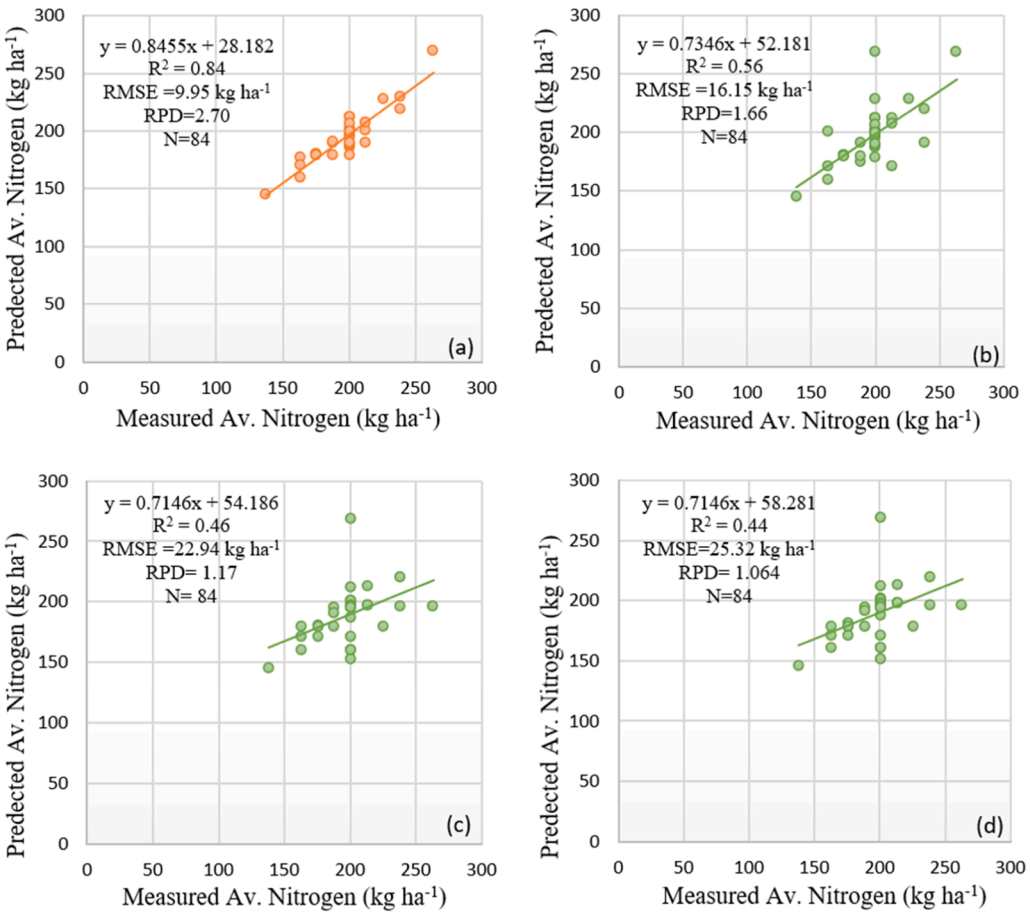


Fig. 6. Mid-infrared (MIR) prediction of soil Available Nitrogen calibration datasets of (a)PLSR (b) RF (c) MARS & (d) SVR Models Respectively.

nitrogen concentrations, offering results comparable to those obtained through conventional analytical methods. Shi et al. (2023) further emphasized that MIR spectroscopy offers more robust prediction across different soil depths compared to Vis–NIR. The present findings corroborate these studies, showing that among the various preprocessing techniques, the Standard Normal Variate (SNV) transformation provided the best model performance. The variable importance patterns were more pronounced in the MIR range, supporting the higher RPD values obtained and confirming MIR as the more suitable spectral region for predicting available nitrogen. Overall, the results support the use of MIR spectra in combination with PLSR modelling and SNV preprocessing as a robust and reliable approach for predicting available nitrogen in Inceptisols soils.

4. Conclusions

The study was carried out in the Indo-Gangetic Plain to assess the efficiency of reflectance spectroscopy in the Vis–NIR and MIR regions for estimating available soil nitrogen, a key determinant of crop productivity and soil fertility. Among the tested models, the Partial Least Squares Regression (PLSR) model demonstrated superior performance values, the MIR spectra provided a better model fit than the Vis–NIR spectra, indicating that the MIR region with high RPD value is more suitable for predicting available nitrogen in the soils of the study area. Thus SNV preprocessing in the MIR range with PLSR algorithm is very well suited to replace conventional method of estimating available nitrogen in the intensively cultivated Indo-Gangetic plain regions of India.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Reports a relationship with that includes:. Has patent pending to. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

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